

Project 4. Graph Algorithms

May 31, 2026

Question 1: Return Correlation

Bounds on ρ_{ij} . The quantity ρ_{ij} is the Pearson correlation coefficient between the log-normalized return time series of stocks i and j , defined as

$$\rho_{ij} = \frac{\langle r_i(t) r_j(t) \rangle - \langle r_i(t) \rangle \langle r_j(t) \rangle}{\sqrt{(\langle r_i(t)^2 \rangle - \langle r_i(t) \rangle^2)(\langle r_j(t)^2 \rangle - \langle r_j(t) \rangle^2)}}.$$

By the Cauchy–Schwarz inequality, the Pearson correlation coefficient always satisfies

$$-1 \leq \rho_{ij} \leq +1.$$

The boundary values are achieved as follows:

- $\rho_{ij} = +1$: the two return series are perfectly positively linearly related, meaning the stocks move in exactly the same direction and in the same proportions at every time step.
- $\rho_{ij} = -1$: the two return series are perfectly negatively linearly related, moving in exactly opposite directions.
- $\rho_{ij} = 0$: there is no linear correlation between the two return series.

Justification for log-normalized returns. We use the log-normalized return

$$r_i(t) = \log(1 + q_i(t))$$

rather than the simple return $q_i(t)$ for the following reasons.

1. **Approximate normality.** For small $q_i(t)$, the approximation $\log(1 + q) \approx q$ holds, but the log transform brings the empirical distribution of returns closer to a Gaussian. Since the Pearson correlation coefficient has the most reliable statistical properties under the normality assumption, using $r_i(t)$ yields more robust correlation estimates.
2. **Time-additivity.** Multi-period log returns are additive over time:

$$\log \frac{p_i(T)}{p_i(0)} = \sum_{t=1}^T r_i(t).$$

Simple returns, by contrast, must be multiplied across periods, making aggregation over different time horizons inconsistent.

3. **Symmetry.** Simple returns are asymmetric: a +50% gain followed by a −50% loss results in a net loss ($1.5 \times 0.5 = 0.75 \neq 1$). Log returns are symmetric around zero, so a +50% gain and a −50% loss approximately cancel, which is economically intuitive.
4. **Variance stabilisation.** The log transform compresses the influence of extreme price movements and reduces heteroscedasticity in the return series, producing a more stationary signal and making the temporal average $\langle \cdot \rangle$ a more meaningful summary statistic.

Question 2: Constructing Correlation Graphs

We compute the edge weight $w_{ij} = \sqrt{2(1 - \rho_{ij})}$ for every pair of the 505 stocks using the daily log-normalized return time series, which spans 216 trading days after alignment on common dates.

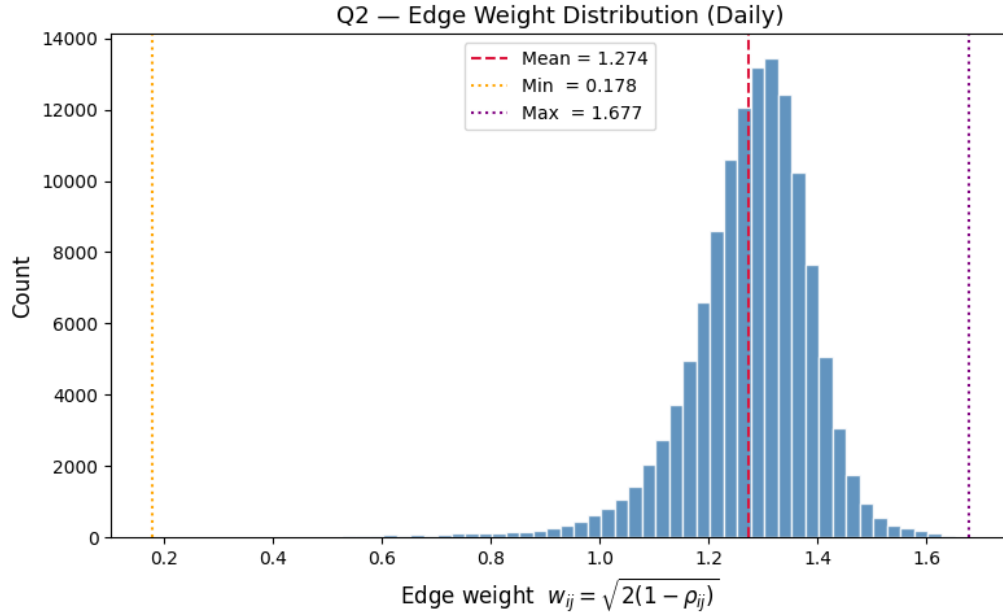


Figure 1: Un-normalized distribution of edge weights w_{ij} for the daily correlation graph.

Statistic	Value
Minimum	0.1779
Maximum	1.6765
Mean	1.2742
Std. dev.	0.1199

Table 1: Summary statistics of the daily edge weight distribution.

The distribution is left-skewed and unimodal, with the bulk of edge weights concentrated between 1.0 and 1.5. A weight of $w_{ij} = 0$ corresponds to $\rho_{ij} = 1$ (perfect positive correlation), $w_{ij} = \sqrt{2} \approx 1.414$ to $\rho_{ij} = 0$ (no correlation), and $w_{ij} = 2$ to $\rho_{ij} = -1$ (perfect negative correlation). The observed mean of 1.274 indicates that the average stock pair has

a mild positive correlation ($\rho_{ij} \approx 0.19$), consistent with the broad market factor driving most S&P 500 constituents in the same direction. The minimum weight of 0.178 corresponds to $\rho_{ij} \approx 0.984$, reflecting nearly co-integrated pairs such as dual-class shares or close industry peers.

Question 3: Minimum Spanning Tree (MST)

We extract the MST of the daily correlation graph using Kruskal’s algorithm, which retains the $|V| - 1 = 504$ edges of lowest weight (highest correlation). The resulting MST has 505 nodes and 504 edges. The highest-degree node is **XEL** (Xcel Energy, Utilities sector) with degree 15, acting as the central hub connecting the Utilities vine cluster to the rest of the tree.

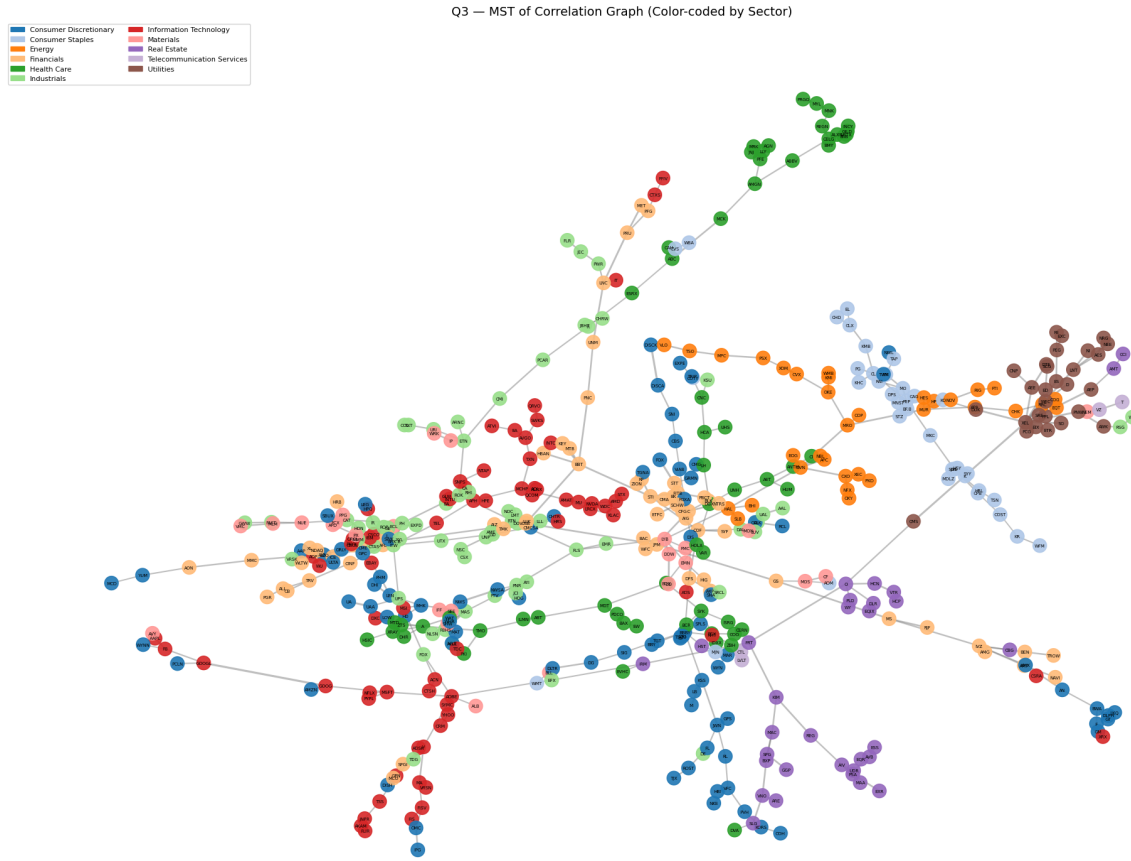


Figure 2: MST of the daily correlation graph, color-coded by market sector. Each node represents a stock and each edge connects the pair with the highest correlation (lowest w_{ij}) in the spanning tree.

Vine cluster pattern. The MST exhibits a clear *vine cluster* structure: stocks belonging to the same sector consistently cluster into elongated, branch-like sub-trees. This pattern arises because intra-sector correlations are systematically stronger than inter-sector correlations —

stocks in the same sector are driven by common economic factors (e.g., oil prices for Energy, interest rates for Financials, consumer spending for Consumer Discretionary), so the MST preferentially connects them first.

Several vine clusters are clearly visible in Figure 2:

- **Utilities** (brown): forms a dense, compact vine in the upper-right region, connected through the hub node XEL.
- **Health Care** (green): occupies a long vine extending toward the top of the graph.
- **Energy** (orange): groups tightly in the right portion of the tree, reflecting the strong common dependence on oil prices.
- **Real Estate** (light blue): forms a separate cluster, consistent with the sector's sensitivity to interest rates rather than broad market movements.
- **Information Technology** (red): dispersed across the central region, reflecting the heterogeneity of the sector (hardware, software, semiconductors).

Hub nodes (high degree) are predominantly large-cap or diversified companies that serve as bridges between sector clusters, confirming that the MST encodes meaningful economic relationships beyond simple market-cap effects.

Question 4: Community Detection on the MST

We apply the Walktrap algorithm (step size = 4) to the daily MST. The algorithm detects **42 communities**. The resulting partition is compared side-by-side against the true sector labels in Figure 3.

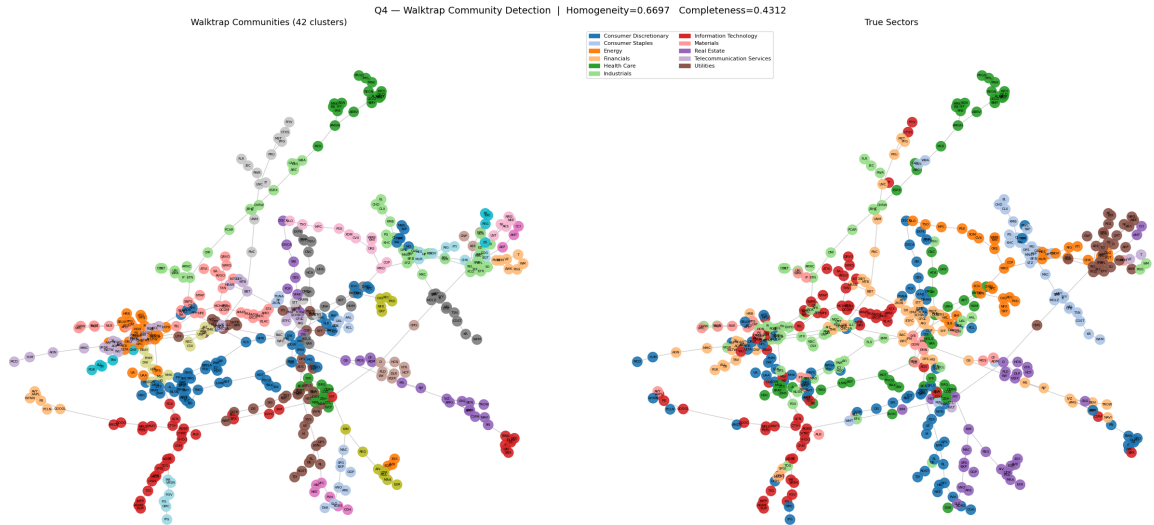


Figure 3: Left: 42 Walktrap communities detected on the daily MST. Right: true sector labels for comparison.

Metric	Value
Number of communities	42
Homogeneity	0.6697
Completeness	0.4312

Table 2: Walktrap community detection metrics for the daily MST.

Interpretation. Homogeneity measures whether each detected community contains only members of a single true sector; completeness measures whether all members of a true sector are assigned to the same community. A homogeneity of 0.67 indicates that most communities are relatively pure — the stocks within a given community tend to belong to the same sector. The lower completeness of 0.43 reflects the fact that Walktrap over-partitions the tree: because the MST vine clusters are long and linear, the random-walk process splits each sector into several smaller communities rather than recovering it as a single unit. This is an inherent limitation of applying graph-based community detection to trees, where every edge is a bridge and the algorithm has no dense subgraph structure to anchor larger communities.

Question 5: Sector Clustering in the MST

We evaluate the sector predictability of the MST using the metric

$$\alpha = \frac{1}{|V|} \sum_{v_i \in V} P(v_i \in S_i),$$

under two definitions of $P(v_i \in S_i)$:

- **Method 1 (MST neighbors):** $P(v_i \in S_i) = |Q_i|/|N_i|$, where Q_i is the set of neighbors of node i in the MST that belong to the same sector, and N_i is the full neighbor set.
- **Method 2 (random baseline):** $P(v_i \in S_i) = |S_i|/|V|$, i.e., the fraction of all stocks that belong to sector S_i , representing the probability of a randomly chosen stock being in the correct sector.

Method	α
MST neighbors	0.8009
Random baseline	0.1146
Ratio	6.99×

Table 3: Alpha metric for the daily MST under two definitions of $P(v_i \in S_i)$.

Interpretation. The MST-neighbor α of 0.80 means that, on average, 80% of a stock’s MST neighbors belong to the same sector. The random baseline of 0.11 reflects the sector composition of the full dataset (11 sectors of varying sizes). The ratio of 6.99× shows that an MST neighbor is nearly 7 times more likely to share the same sector than a randomly selected stock, confirming that the MST captures strong sector structure and would be a useful tool for predicting the sector of an unknown stock from its neighbors.

Question 6: Correlation Graphs for Weekly Data

We repeat Questions 2–5 using weekly data sampled on Mondays, yielding 38 observations per stock after alignment on common dates.

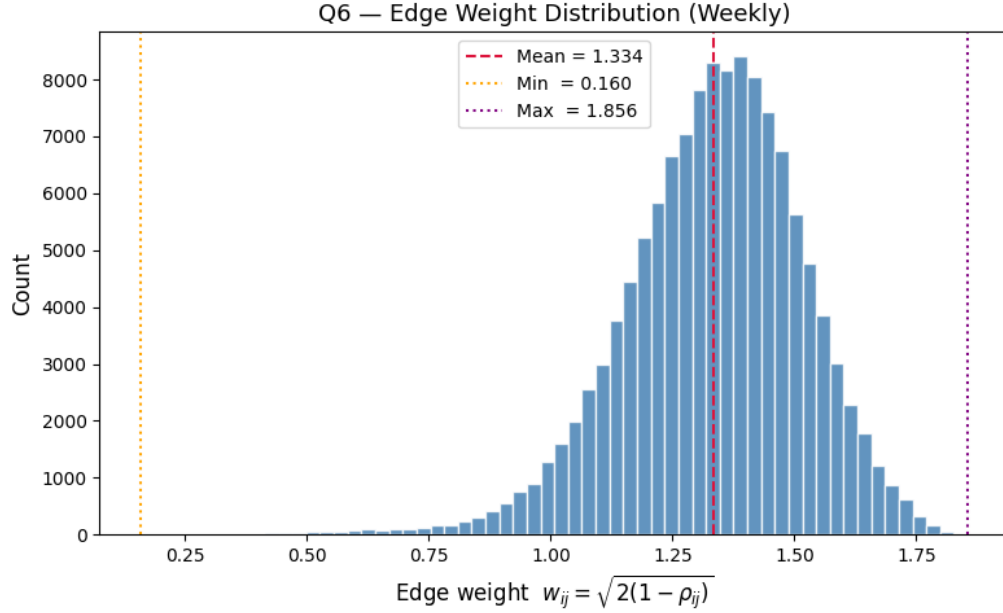


Figure 4: Un-normalized distribution of edge weights w_{ij} for the weekly correlation graph.

Statistic	Daily	Weekly
Minimum	0.1779	0.1596
Maximum	1.6765	1.8556
Mean	1.2742	1.3335
Std. dev.	0.1199	0.1839

Table 4: Edge weight statistics: daily vs. weekly.

Q2 (Weekly) — Edge Weight Distribution. Compared to daily data, the weekly distribution is wider (higher std. dev. 0.184 vs. 0.120) and shifted slightly rightward (higher mean 1.334 vs. 1.274), indicating that weekly correlations are on average weaker and more dispersed. This is expected: averaging over a full week dampens short-term co-movements, reducing intra-sector correlation and broadening the weight distribution.

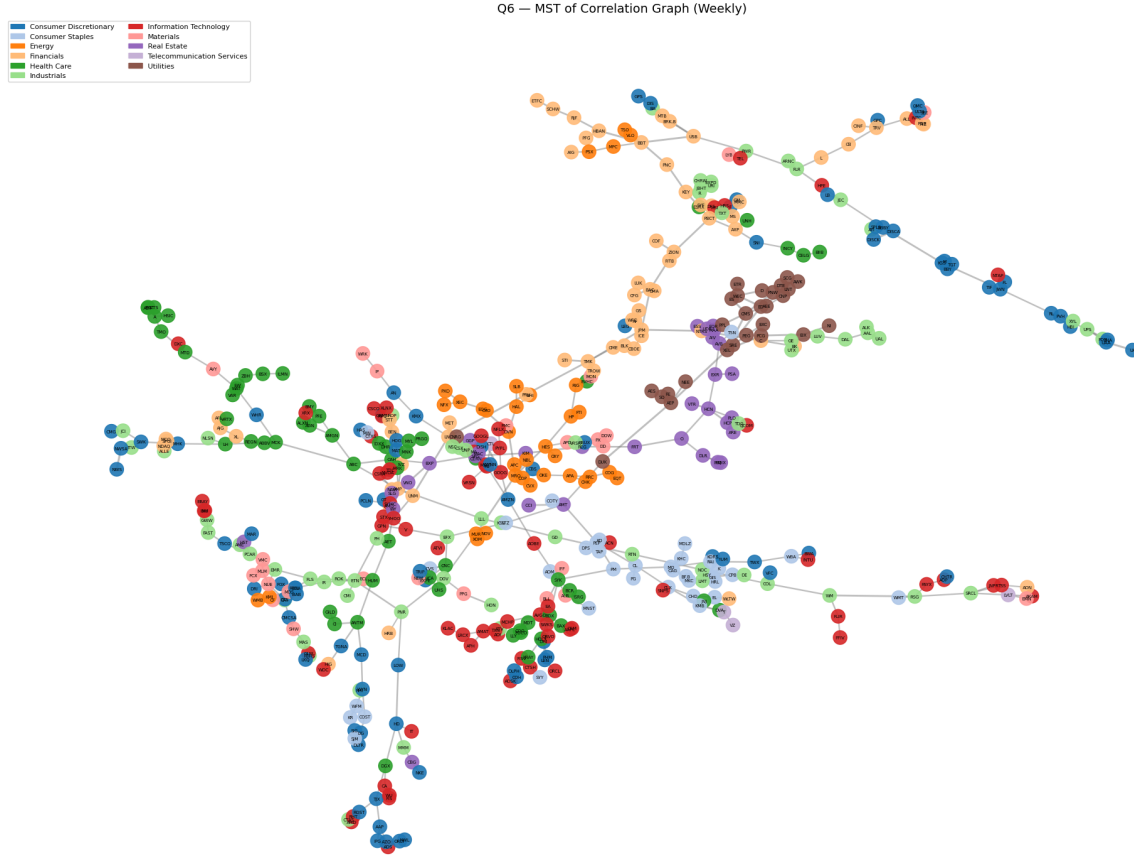


Figure 5: MST of the weekly correlation graph, color-coded by sector. Highest-degree hub: AMP (Financials, degree 8).

Q3 (Weekly) — MST. The weekly MST retains the vine cluster structure observed in the daily MST, with same-sector stocks grouping into elongated branches. However, the topology is visibly different: the hub node changes from XEL (Utilities, degree 15) in the daily MST to AMP (Financials, degree 8), and the maximum degree drops, indicating a flatter, less star-like tree. Sector clusters are less compact, reflecting the weaker and noisier weekly correlations.

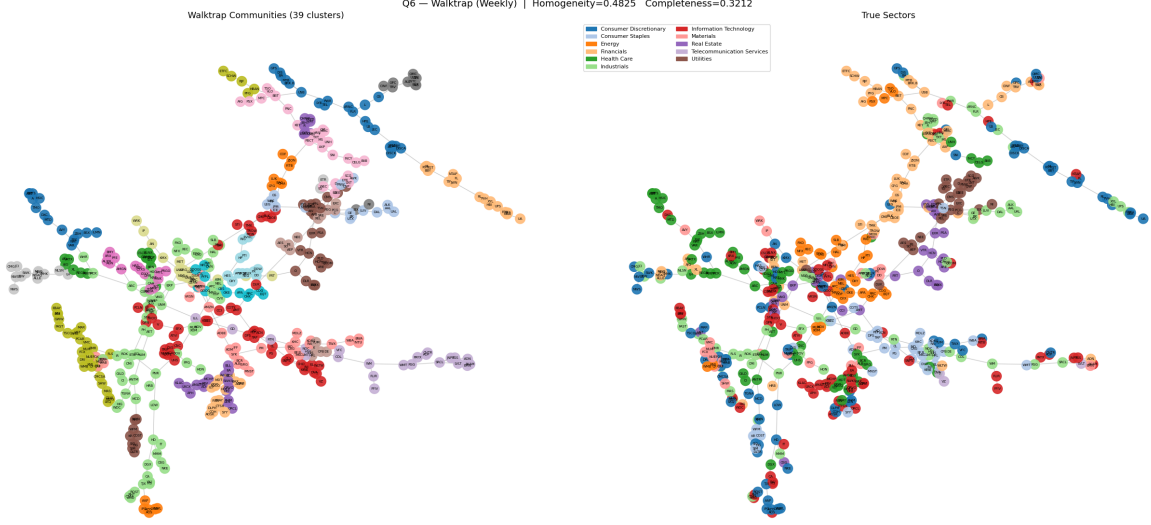


Figure 6: Left: 39 Walktrap communities on the weekly MST. Right: true sector labels. Homogeneity = 0.4825, Completeness = 0.3212.

Metric	Daily	Weekly
Communities	42	39
Homogeneity	0.6697	0.4825
Completeness	0.4312	0.3212

Table 5: Walktrap metrics: daily vs. weekly MST.

Q4 (Weekly) — Walktrap Community Detection. Both homogeneity and completeness drop substantially from daily to weekly, confirming that community detection becomes less reliable with fewer data points. With only 38 weekly observations, the correlation estimates carry higher variance, causing the MST to include more spurious cross-sector edges that fragment the vine clusters and confuse the Walktrap algorithm.

Method	Daily	Weekly
MST neighbors	0.8009	0.6078
Random baseline	0.1146	0.1146
Ratio	6.99×	5.31×

Table 6: Alpha metric: daily vs. weekly MST.

Q5 (Weekly) — Alpha Metric. The MST-neighbor α falls from 0.80 (daily) to 0.61 (weekly), and the ratio drops from 6.99×

Question 7: Correlation Graphs for Monthly Data

We repeat Questions 2–5 using monthly data sampled on the 15th of each month, yielding only 6 observations per stock after alignment on common dates.

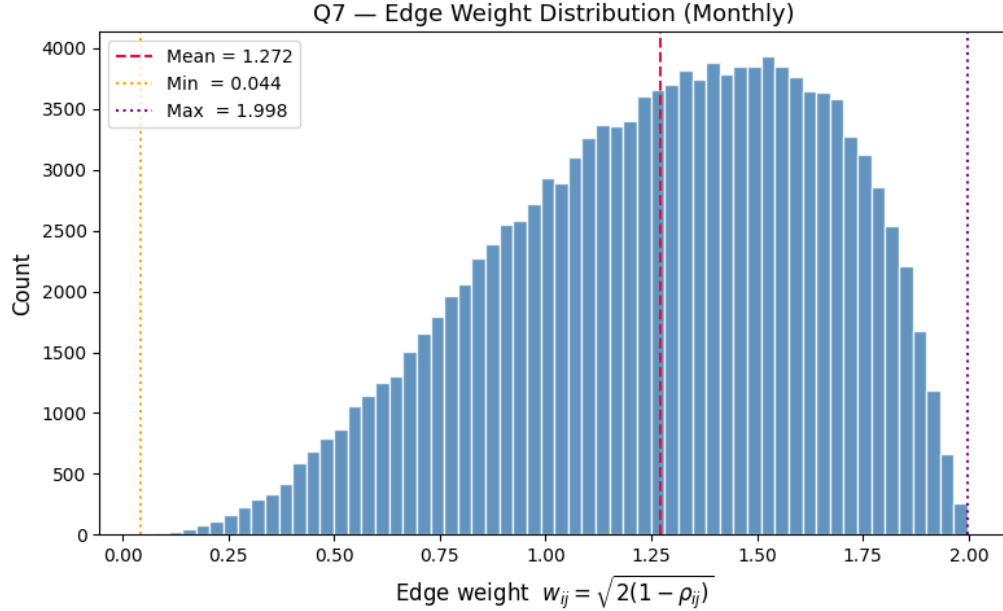


Figure 7: Un-normalized distribution of edge weights w_{ij} for the monthly correlation graph.

Statistic	Daily	Weekly	Monthly
Minimum	0.1779	0.1596	0.0435
Maximum	1.6765	1.8556	1.9976
Mean	1.2742	1.3335	1.2720
Std. dev.	0.1199	0.1839	0.3836

Table 7: Edge weight statistics across all three granularities.

Q2 (Monthly) — Edge Weight Distribution. The monthly distribution is dramatically wider (std. dev. 0.384, more than three times the daily value) and nearly uniform across the full range $[0, 2]$, approaching the theoretical limits of $w_{ij} \rightarrow 0$ ($\rho_{ij} \approx 1$) and $w_{ij} \rightarrow 2$ ($\rho_{ij} \approx -1$). This is a direct consequence of the very small sample size of only 6 monthly observations: with so few data points, the Pearson correlation estimates are highly unreliable and can take extreme values by chance, producing a near-uniform edge weight distribution that carries little economic signal.

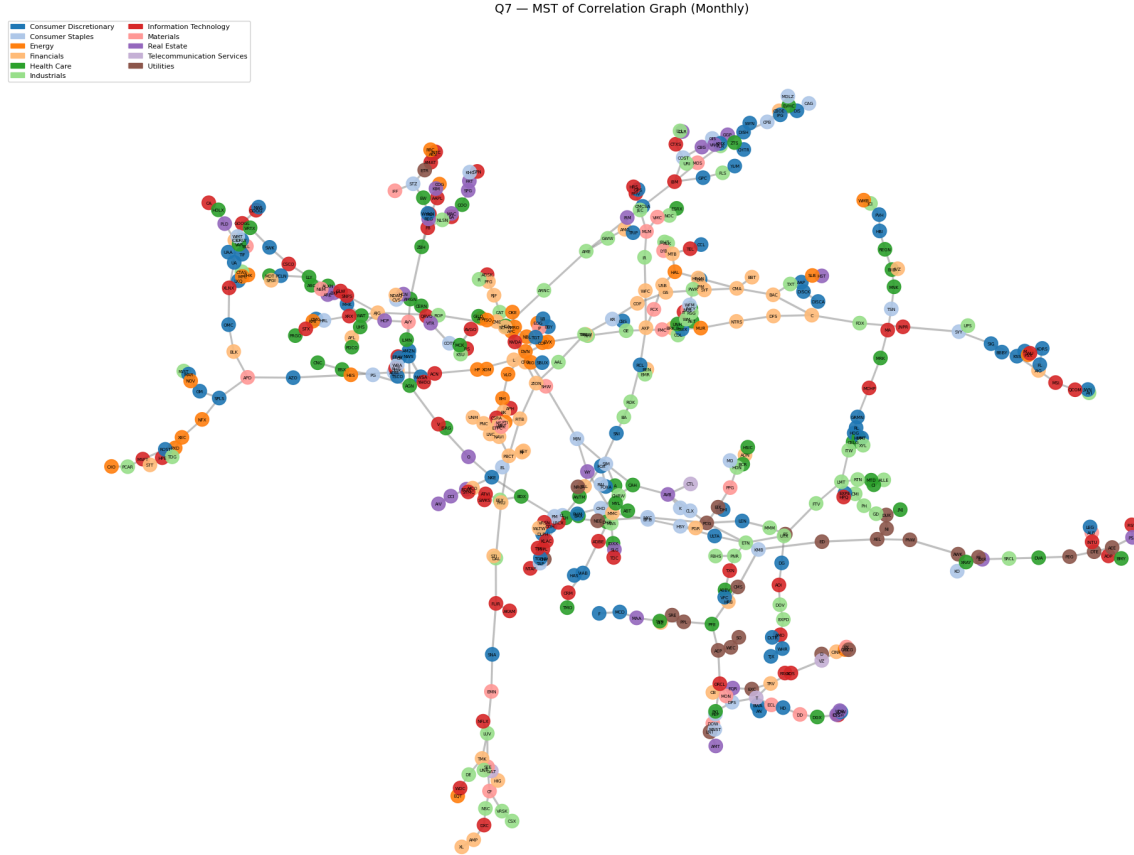


Figure 8: MST of the monthly correlation graph, color-coded by sector. Highest-degree hub: LMT (Industrials, degree 5).

Q3 (Monthly) — MST. The monthly MST shows a markedly more diffuse structure compared to the daily and weekly MSTs. The vine clusters are largely absent: same-sector stocks are scattered across the tree rather than grouped into distinct branches. The hub node is LMT (Lockheed Martin, Industrials) with degree only 5, the lowest of the three granularities, indicating a flatter and less centralized tree.

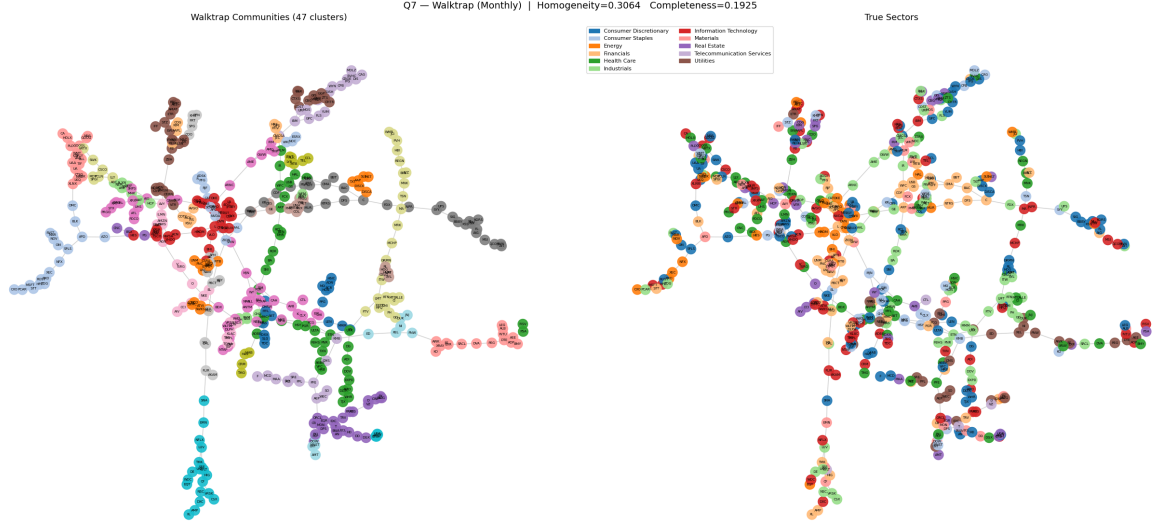


Figure 9: Left: 47 Walktrap communities on the monthly MST. Right: true sector labels. Homogeneity = 0.3064, Completeness = 0.1925.

Metric	Daily	Weekly	Monthly
Communities	42	39	47
Homogeneity	0.6697	0.4825	0.3064
Completeness	0.4312	0.3212	0.1925

Table 8: Walktrap metrics across all three granularities.

Q4 (Monthly) — Walktrap Community Detection. Both homogeneity and completeness reach their lowest values at monthly granularity. The 47 detected communities are largely meaningless from a sector perspective, reflecting the noise-dominated correlation estimates that arise from only 6 data points.

Method	Daily	Weekly	Monthly
MST neighbors	0.8009	0.6078	0.2238
Random baseline	0.1146	0.1146	0.1146
Ratio	6.99×	5.31×	1.95×

Table 9: Alpha metric across all three granularities.

Q5 (Monthly) — Alpha Metric. The MST-neighbor α collapses to 0.22 at monthly granularity, barely above the random baseline of 0.11 (ratio 1.95×). This confirms that with only 6 monthly observations, the MST captures almost no reliable sector information and would be a poor tool for predicting the sector of an unknown stock.

Question 8: Comparison of Daily, Weekly, and Monthly Data

Table 10 consolidates all metrics computed in Questions 2–5 across the three temporal granularities.

Metric	Daily	Weekly	Monthly
Observations per stock	216	38	6
Edge weight mean	1.2742	1.3335	1.2720
Edge weight std. dev.	0.1199	0.1839	0.3836
Walktrap communities	42	39	47
Homogeneity	0.6697	0.4825	0.3064
Completeness	0.4312	0.3212	0.1925
α (MST neighbors)	0.8009	0.6078	0.2238
α (random baseline)	0.1146	0.1146	0.1146
α ratio	$6.99\times$	$5.31\times$	$1.95\times$

Table 10: Summary of all metrics across daily, weekly, and monthly granularities.

What changes across granularities? Every quality metric — homogeneity, completeness, α , and α ratio — decreases monotonically as the sampling frequency decreases from daily to weekly to monthly. The edge weight standard deviation increases dramatically (from 0.12 to 0.38), reflecting increasingly unreliable correlation estimates as the number of observations shrinks. The monthly histogram approaches a near-uniform distribution over $[0, 2]$, indicating that with only 6 data points the Pearson estimator is essentially noise. The MST topology also changes: the daily hub (XEL, degree 15) gives way to a lower-degree hub at weekly (AMP, degree 8) and an even flatter tree at monthly (LMT, degree 5), consistent with correlations becoming weaker and less structured.

What remains similar? The α ratio remains above 1 at all three granularities, confirming that MST neighbors are always more likely to share the same sector than a randomly chosen stock. The broad vine cluster pattern is partially visible even at weekly granularity. The mean edge weight stays close to 1.27–1.33 across all granularities, suggesting that the average pairwise correlation level is relatively stable; it is the *reliability* of individual estimates, not their average, that degrades with coarser sampling.

Which granularity gives the best results for sector prediction? Daily data gives the best results by a clear margin. With 216 observations, the Pearson correlation estimates have low variance, producing a reliable MST that faithfully reflects true sector structure. This is evidenced by the highest α ratio ($6.99\times$), homogeneity (0.67), and completeness (0.43). Weekly data is a moderate compromise: the 38 observations reduce noise compared to monthly, but still produce noticeably weaker sector clustering than daily. Monthly data with only 6 observations is effectively unusable for sector prediction — the α ratio of $1.95\times$ is barely above the random baseline, and the MST captures almost no sector information.

The fundamental reason is statistical: the Pearson correlation estimator requires a sufficient number of observations T to achieve low variance. For $T = 6$, the standard error of $\hat{\rho}$ is on the order of $(1 - \rho^2)/\sqrt{T} \approx 0.4$, which is large enough to make even the strongest sector

correlations indistinguishable from noise. Increasing T to 216 (daily) reduces this error by a factor of $\sqrt{216/6} = 6$, yielding correlation estimates that reliably rank intra-sector pairs above inter-sector pairs and produce meaningful MST structure.